

Reconnecting With Nature

EcoNest

IoT Challenge: End-to-end system solution for the Internet of Things

Gotzon Viteri - 79077252D
Josu Viteri - 79077253X
Markel Mencía - 79135145B
Asier Aurre - 72230207H

October 2026

2026 - Desarrollo de Aplicaciones para Internet de las Cosas

1 Description of the IoT system and its purpose

Bilbao is a city with several parks and green spaces. While the infrastructural well-being of these places is already maintained by the extensive cleaning service team of the city, it is difficult to manage the animals that live in public spaces, which are mostly pigeons.

Since these animals can pose a threat to certain demographics, due to their ability to infect humans with parasites or illnesses, it's important to monitor pigeons to ensure public natural spaces are safe. Moreover, pigeons are very efficient at finding food, which can affect the ecosystem of other animals in parks or natural spaces. Making sure pigeon overpopulation doesn't happen in Bilbao is key to maintaining a healthy natural environment, which could in turn encourage people to enjoy these spaces more than they do already.

We propose an intelligent, value-driven Internet of Things system designed to remotely monitor public spaces with sensors and machine learning algorithms designed to identify birds, especially pigeons. With its detection capabilities, it will be easier to locate pigeons and identify their patterns. This not only can help city officials track their behaviour and make appropriate decisions, but also help citizens be conscious of the behaviour of pigeons too, as we intend to make this project not only available to the city hall, but also the general public.

2 Use case scenario

The main use cases this solution offers is the monitoring it provides to the municipality. When a citizen makes a report (which is out of scope for this project and is already managed by the Bilbao city hall), the city staff would have a quick way to monitor the situation in the area with our solution. Similarly, the citizens could access the platform for the same thing. The following are our proposed use cases:

- Display pigeon count in an area
- Display heatmaps
- Show pigeon history in an area
- Display screenshots

3 Advantages over market alternatives or existing systems

Bilbao already has plans and protocols in place to regulate and monitor pigeons around public spaces in the city [1]. Citizens may report incidents to said program. However, the protocol they carry out is slow and expensive. When a report is received, a team of officials is sent to inspect the area. Then, a report is written about the situation, and if they see it fit, action is taken, whatever it is. The issue is that moving personnel to a space can be very time-consuming and costly, when there may be better alternatives.

4 Solution architecture diagram

Figure 1 represents the proposed solution architecture diagram for this project.

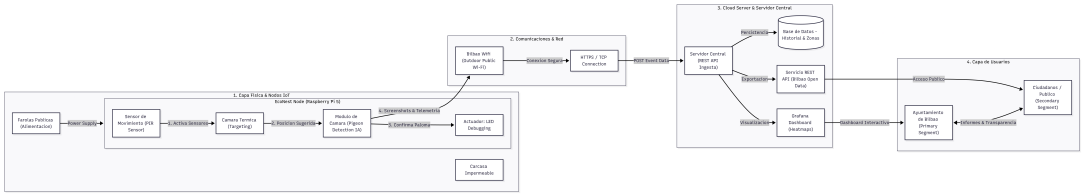


Figure 1: Solution architecture diagram

5 Sensors and actuators

5.1 Sensors

- **Motion sensor**: It will be the initial sensor. When it detects movement, the other sensors will turn on. This provides a huge save in electricity, as some of the other sensors are not lightweight.
- **Thermal camera**: The average temperature of a pigeon (or other birds) is slightly higher than humans'. A thermal camera can help identify the position of a pigeon with its temperature.
- **Camera module**: Used as the input for a pigeon recognition model. It works along with the thermal camera to focus its activity on the suspected position of the pigeon. Once it detects one, it takes a screenshot.

5.2 Actuators

Since this solution mostly works online and works as a passive device, physical actuators are not really needed. However, a debug LED could be implemented to turn on if the device is currently detecting a pigeon.

6 Physical/link layer communication technologies

6.1 Power

Since these devices are meant to be placed in parks, they would be easy to integrate in streetlights. We propose the use of Raspberry Pis due to their computing power and low digital footprint. They could use the same energy streetlights receive without much hassle. Microcontrollers are meant to be able to be executed in a low-power environment.

6.2 Environmental resistance

The devices would need to be enclosed in a waterproof case that would cover the controller and its sensors.

6.3 Communication

Bilbao already offers outdoor Wi-Fi connectivity with Bilbao Wifi. Integrating these devices to it would not be difficult, and it would suffice to connect all devices to the central server. From there, it would be served to the citizens and the city staff.

7 Application/data layer communication technologies

The devices would send the data through a secure HTTPS connection over TCP, as per the standard on the internet. MQTT could be used, but since there is no connection between monitoring devices and Raspberry Pi devices can easily withstand HTTPS connections, the traditional HTTPS protocol would suffice. It is important to keep in mind that these devices do not work as recurrent video devices, but as detection systems that would only send data when they have detected a pigeon.

8 Cloud-based monitoring dashboard and data analytics

Grafana is a very good tool to display the data gathered by the devices in a centralized and visual way. It would run on the main server for everyone to explore. Data could also be acquired from the central server with a REST API architecture implementation.

As for external data, it would not be necessary for it to use any. Using it would increase the energy intake, and the data gathered from its on-chip sensors is enough for the use cases it serves. However, it could be a provider for Bilbao Open Data [2]. The central server could parse the data to an appropriate format to then send to its database.

9 Ethics, privacy and environmental impact

Since our IoT system involves monitoring, data transparency is quite important. Our product is not intended and will not be used for any form of surveillance. Its use cases are limited to gathering statistical information about pigeons and other birds. To ensure this, the source code and the models it will use will be open-source, and of course, all the data gathered will not only be available to the city hall, but also the public.

The environmental impact is also important to be addressed. Running programs like these can be quite cheap if you use the proper hardware. Raspberry Pi devices turn out to be very efficient to run recognition models like this, while also having a very small energy footprint. That is why we choose Raspberry Pi 5 devices for this project.

As for keeping screens out of nature and encouraging people to be in touch with it more regularly, our project can help by the information we provide with it. Citizens will know what places to avoid and city personnel will know where to look to keep the public spaces safe. Making sure our parks are environmentally stable is key to making them more attractive and immersive, and pigeon overpopulation is a roadblock for that, as it can affect the biodiversity of these spaces.

10 Business model

Our proposition is EcoNest, a remote, real-time access to Bilbao's public spaces' pigeon presence, so that decisions are based on data rather than on slow on-site inspections. Figure 10 shows the business model canvas for the solution.

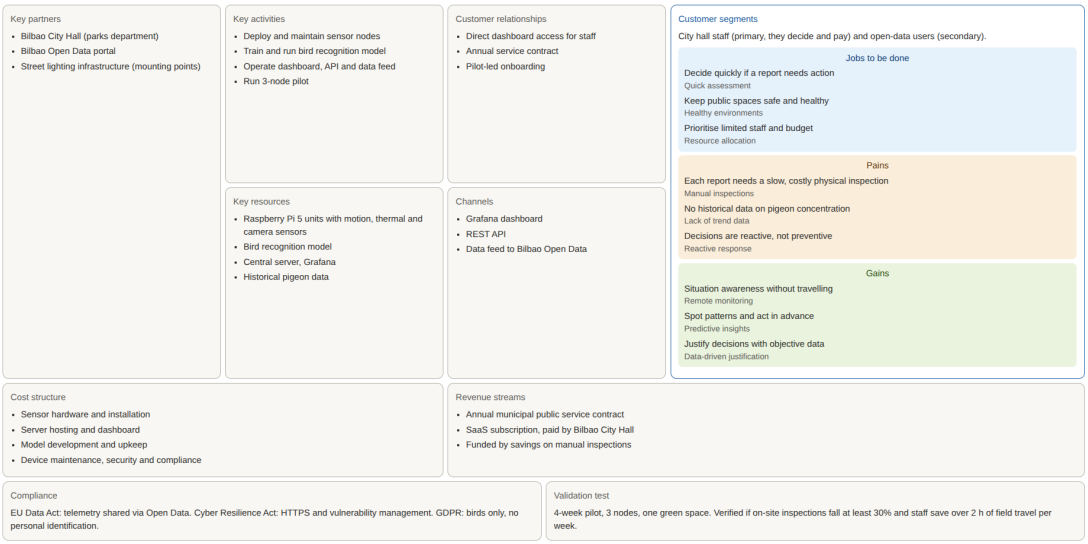


Figure 2: Business Model Canvas

11 Value Proposition

In solutions like these, it is important to highlight where the value is. The issue of the Internet of Things must be justified. For this, we propose the following value proposition canvas:

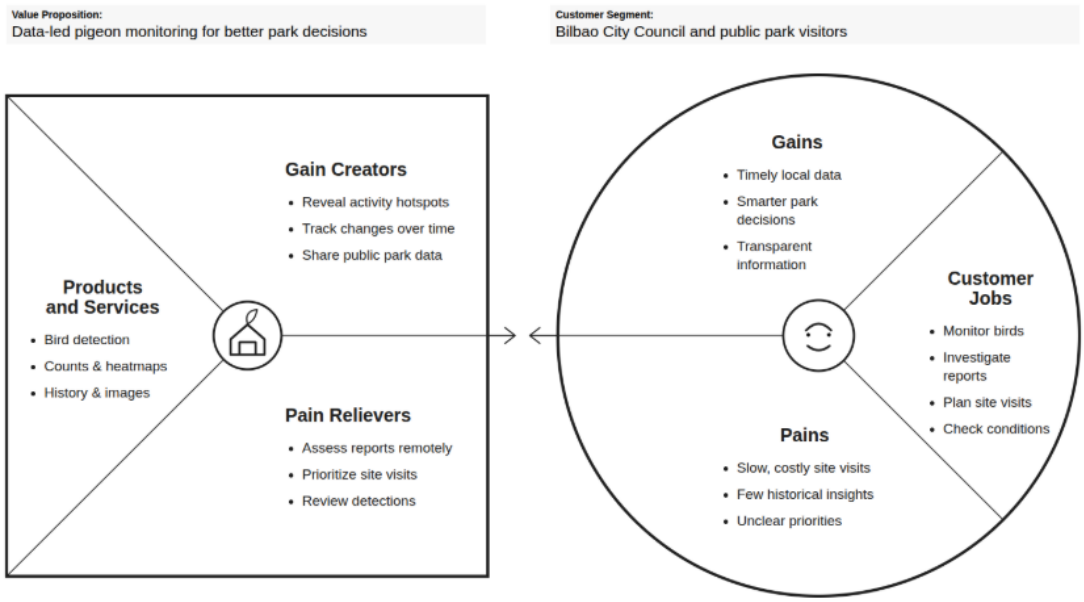


Figure 3: Value Proposition Canvas

12 Initial task planning

In order to provide a proof of concept for the project, we will build an MVP (Minimum Viable Product), which serves to deliver value for the hypothesis of our project by developing a product offering a minimal implementation that serves the purpose of the solution. In order to build it, the Scrum [3] methodology will be used, using four, one-week sprints:

- **Sprint 1**: Initial Raspberry Pi setup, camera acquisition
- **Sprint 2**: Pigeon recognition model implementation
- **Sprint 3**: Central server development: Grafana and REST API
- **Sprint 4**: Raspberry Pi and central server connection development

References

[1] Bilbao City Council. *Prevención y control de la población de aves en el municipio*. Available at: `https://www.bilbao.eus/cs/Satellite?c=BIO_Servicio_FA&cid=1279195977419&language=es&pageid=3000070319&pagename=Bilbaonet%2FBIO_Servicio_FA%2FBIO_Servicio`

[2] Bilbao City Council. *Bilbao Open Data Portal*. Available at: `https://www.bilbao.eus/opendata/es/open-data`

[3] Agilismo en Ingeniería de Software. *Scrum, Desarrollo Ágil: Desarrollo mediante MVP*. (2016). Available at: `https://agilismoeningenieriadesoftware.blogspot.com/2016/08/scrum-desarrollo-agil-desarrollo.html`